SERGIO VERDUGO RUBIO

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SUMMARY

Computer Science and Artificial Intelligence student at IE University with strong problem-solving skills and hands-on experience building data and machine learning pipelines on AWS and PySpark for location-intelligence products. Proven track record of improving performance and reducing costs by optimising pipelines and working within tight technical and business constraints. Highly driven and hard-working, with a strong ambition to keep growing personally and professionally while taking on challenging engineering problems and contributing to AI-powered products across different industries.

EDUCATION

IE UNIVERSITY **Madrid, Spain**

Bachelor's in Computer Science and Artificial Intelligence **Expected: May 2026**

- Current GPA: 8.2/10.00 - IE High Potential Award Scholarship, IE Foundation Fellow.

BRITISH SCHOOL OF MALAGA **Malaga, Spain**

Advanced Levels (A Levels) – Physics (A), Mathematics(A*), Further Mathematics(A*). **Graduated: June 2022**

EXPERIENCE

MEmob+ – Data & Location Intelligence (AdTech) **Remote (Dubai, UAE)**

Machine Learning Engineer Intern **June 2025 – December 2025**

- Built scalable data pipelines on AWS EMR + PySpark to process large-scale mobility data into production-ready datasets for MEmob+'s location-intelligence products. Designed an "Enriched User Profiles" dataset linking anonymized users to Points of Interest and behavioural features, enabling downstream traffic analysis and audience insights.
- Prototyped a road-heatmap / route-mapping module using the Mapbox Directions API and geospatial optimisation, reducing required external API calls by ~95% and staying under strict call limits.
- Identified a vulnerability in the data ingestion process where over 75% of the location data was fraudulent and was resulting in significant losses for the company.

Blockchain Startups – Bproto Labs & Cortex Network **Remote (Colorado, USA)**

COO, Business Developer and Co-founder **April 2021 – May 2024**

- Conducted market research and secured funding to develop blockchain and AI software solutions, including a grant from Aleph Zero to build a decentralized DNS protocol in Rust. Developed and executed strategic plans that doubled the size of the company.
- Played a key role in fundraising (~$1.5M in venture capital) by translating technical roadmap and market analysis into investor-ready materials.

Brainy Buddy **Remote (Madrid, Spain)**

Founder and Engineer **January 2026 – Current**

- Designed and built BrainyBuddy, an AI-powered study planner, owning product vision, UX design, and full-stack implementation (Next.js, FastAPI, PostgreSQL) from zero to production.
- Implemented end-to-end AI solutions including adaptive scheduling with constraint optimization, exam generation with professor style imitation, spaced repetition flashcards, and LLM-powered tutoring (LangGraph + Gemini).

EXTRACURRICULAR ACTIVITIES

- **UBRI Connect & XRP Ledger Apex (Singapore)**: Selected as one of three students to represent IE University at Ripple's UBRI Connect academic conference and XRP Ledger Apex Summit in Singapore; fully funded trip including participation in the official XRPL hackathon (Jun 2025).
- **Money 404NotFound:** Developed the Frontend, Backend and Infrastructure of a Banking app that included features such as user registration and administrator privileges. The app was developed with Flask, using Vue.js for the front-end and IaC with GitHub actions to deploy everything to Azure (Dec 2024).
- **Stratego AI Player:** Created an AI agent that played Stratego using a combination of minimax with alpha-beta pruning using different heuristics to evaluate the best move in every board state. (Nov 2024)
- **Eternal – WhatsApp Conversation Preservation Web App**: Built and launched a full-stack app using Django, Vue.js, PostgreSQL, Google Auth and Celery to archive and visualise WhatsApp conversations; gained end-to-end product and deployment experience while undergoing a full-time internship to complete experience gaps with other technology (Nov 2025).

ADDITIONAL INFORMATION

- **Languages: Spanish (Native), English (Bilingual), German (Basic-Intermediate), French (Basic).**
- **Technology Stack:**
  - **Languages**: Python(professional), C++(intermediate), SQL(professional), C(intermediate), JS(intermediate).
  - **Frameworks and libraries:** Python, C++, C, SQL, PySpark, pandas, scikit-learn, PyTorch, Django, Flask, Vue.js, PostgreSQL, MongoDB, AWS (EMR, S3), Azure, Docker, Celery, GitHub Actions, LangGraph

Interests: Football, Watersports (worked as a windsurf instructor in the Summer of 2020), Economics, Finance, Resource Optimization.